SENSITIVITY OF AN AUDIO-DEEPFAKE LEADERBOARD TO THE RESAMPLING UNIT AND TO SOURCE, SPEAKER, AND ATTACK WEIGHTING

Rafaello Virgilli, Lucas Alcantara Souza, Alexandre Costa Ferro Filho, Daniel Tunnermann, Gustavo dos Reis Oliveira, and Arlindo Galvão Filho

Universidade Federal de Goiás, Brazil
rafaello.virgilli@discente.ufg.br

ABSTRACT

Audio-deepfake leaderboards rank detectors from repeated trials that share speakers and spoofing attacks. We analyze eight systems on ASVspoof 2021 DF, giving 28 pairwise contrasts. For systems A and B, $\Delta\mathrm{EER}_{A,B} = 100(\mathrm{EER}_A - \mathrm{EER}_B)$ percentage points; a pair is separated when its 95% simultaneous max-t band excludes zero. A trial-level bootstrap separates 26/28 pairs and 5/6 organizer-baseline pairs, whereas a bootstrap that resamples speaker and attack identities independently separates 18/28 and 0/6. On single-source SpoofCeleb, the same comparison on a different four-detector roster changes from six of six pairs separated to three, showing that between-source mixture changes are not necessary. Under twelve evaluation-weighting rules defined over source, speaker and attack groups, six of the twelve within-cohort pairs reverse their point ordering and none of the sixteen between-cohort pairs does. All sixteen pairs between the four SSL detectors and the four organizer baselines remain separated under both bootstraps, each leave-one-source-out jackknife refit and every weighting rule, whereas within-cohort separation changes with the resampling procedure and with source deletion, and within-cohort point ordering reverses under some weighting rules. These are fixed-score sensitivity bands, not population confidence intervals.

Index Terms— Anti-spoofing, audio deepfake detection, evaluation methodology, clustered bootstrap, ranking uncertainty

1. INTRODUCTION

In 2021 the ASVspoof organizers tested every pair of submitted systems for significance, applied a Holm–Bonferroni correction, and published the deepfake task's systems in four groups, noting that the top two were not separated [1]. We found no uncertainty analysis in three subsequent benchmark reports: the journal overview [2], ASVspoof 5 [3], and the recent multi-benchmark leaderboard [4] contain none of *significance test*, *confidence interval*, *bootstrap*, *p-value* or *standard error*. The sub-point margins on that modern leaderboard are therefore reported without an uncertainty test; a 2026 benchmark of self-supervised models likewise ranks point EERs without an interval, error bar or test [5]. The ASVspoof 5 overview [3] also studies score calibration; calibration establishes whether a score means what it says, not whether two systems differ.

That matters more than it would if the analyses the field did run had represented the dependence. ASVspoof 2017 reported 95% EER confidence intervals and read their overlap [6], and 2021 tested every pair [1]; both use [7]. The three later benchmark reports named above contain no such analysis, although bootstrap EER spreads [8] and resampled rank frequencies [9] do appear in recent detector and benchmark papers. Both variants of the cited test [7] treat trials within a class as independent Bernoulli draws; the dependent form accounts for two systems sharing trials, not for trials sharing speakers and attacks. ASVspoof 2021 DF contains 533,928 evaluation trials but only 93 speakers and 110 attacks, and per-attack EERs for SSL-AASIST span 0.4–10.7%. Across the eight detectors, the one-way random-intercept ICC $\sigma_s^2/(\sigma_s^2 + \sigma_e^2)$ on bona-fide scores ranges from 0.15 to 0.77, where σ_s^2 and σ_e^2 are the speaker-intercept and residual variances. ICC alone does not establish dependence in paired EER differences; below we measure how paired contrasts respond to two declared perturbation laws on the joint scores, and repeat the comparison on two external datasets.

We audit two evaluation choices on fixed released scores: whether trials or speakers and attacks are resampled, and how observed source, speaker and attack groups are weighted. The primary question is whether pairwise separation survives those choices; we propose no new bootstrap and claim no identified population sampling law. On 21DF the six organizer-baseline pairs change from five separated to none, and on single-source SpoofCeleb the corresponding four-detector comparison changes from six to three. Alternative positive weighting rules then localize the stable conclusion: separation between the SSL-detector and organizer-baseline cohorts survives, whereas within-cohort ordering does not.

Scope. The primary set comprises eight score files released by the corresponding model authors or ASVspoof organizers, not the universe of public re-scores. We analyze all eleven Speech DF Arena score files for the same trials [4] separately. Four VCC spoof strata contain only 4/4/4/6 speakers, so no 21DF separation result transfers to new speakers or attacks. The bands and reweighting results describe fixed scores under declared procedures, not population confidence or safety.

2. RELATED WORK

This benchmark's own reporting. ASVspoof 2021 Fig. 4(c) pairs systems with the Bengio–Mariéthoz test [7] under Holm correction and publishes group structure [1]; Wang and Yamagishi [10] showed that changing only the training seed moves ADD rankings—a variance component our intervals do *not* include.

Clustered resampling. Poh, Martin and Bengio [11] showed that biometric uncertainty is governed by users rather than sample count; Wu et al. carried user-aware resampling into speech [12] and developed a paired correlation-corrected two-system test [13]. Bolle et al. proposed a subsets bootstrap [14], while Fogliato et al. found undercoverage for dyadic 1:1 matching [15]. Our trials instead have one speaker and, for spoof, one attack: two crossed factors [16], a delicate design [17, 18] distinct from nested metrics [19]. Speech bootstrap intervals also include utterance- and block-based forms [20, 21]. Our contribution is the audited fixed-data result: after EER threshold refitting and all-pair multiplicity are aligned, outputs change with the declared perturbation unit. We claim neither cell-level agreement with Fig. 4(c) of the ASVspoof 2021 report nor population confidence.

Evaluation-design audits. Zhuang et al. [22] compute a paired-binomial detectable effect for quantization benchmarks, treating items as independent—the assumption under test here. In audio deepfakes, aggregation weights change conclusions [23], channel presentation can dominate model choice [24], and other audits question representations [25] and corpus overlap [26]. Speaker-conditioned detector-score sensitivity has also been probed through identity perturbations [27]; our endpoint is paired EER under resampling and evaluation reweighting. None studies pairwise ranking changes under both dependence-aware resampling and alternative source/speaker/attack weights. Simultaneous rank intervals are established [28]; adjacency is score-selected, so we control multiplicity over all 28 contrasts.

3. METHOD

3.1. EER contrast and separation event

Two deterministic systems on a fixed trial list have no sampling uncertainty until a perturbation law is declared. Scores are oriented so larger means bona fide and sorted in ascending order. At each ordered position, bona-fide mass at or below it is the false-reject rate (FRR) and spoof mass above it is the false-accept rate (FAR); the first position minimizing $|\mathrm{FRR}-\mathrm{FAR}|$ is selected and EER is their mean, without interpolation. A distinct-score-boundary recomputation changes no displayed value by 0.001 point. For ordered pair (A, B), $\Delta\mathrm{EER}_{A,B} = 100(\mathrm{EER}_A - \mathrm{EER}_B)$ percentage points (negative favors A); every replicate refits both thresholds. A pair is *separated under procedure* h when its simultaneous band excludes zero; write this indicator as $I_{p,h}$. The identified object is the change in $\Delta\mathrm{EER}$ and $I_{p,h}$ on the fixed scores, not population uncertainty.

3.2. Trial and speaker×attack bootstraps

The trial bootstrap independently resamples trials within class. In the speaker×attack product-weight (PW) bootstrap [18], speakers and spoof attacks are independently resampled with replacement; a spoof trial receives the product of its speaker and attack multiplicities, while a bona-fide trial receives only its speaker multiplicity. Both procedures use the same scores, threshold rule, B=5000 shared draws and all-pair multiplicity correction.

3.3. All-pair multiplicity for procedure bands

For procedure h, let $s_{p,h} = \mathrm{sd}_b(\Delta^{(b)}_{p,h})$. We set $q_{.95,h}$ to the empirical 95th percentile of $\max_{p=1}^{28} |\Delta^{(b)}_{p,h} - \bar{\Delta}_{p,h}|/s_{p,h}$ and report $\hat{\Delta}_p \pm q_{.95,h} s_{p,h}$ [29]. The maximum runs over all 28 contrasts, so each procedure yields one simultaneous band construction; no population error-control guarantee is asserted. Because a population sampling law is not identified for 21DF, we call these procedure-sensitivity bands, not confidence intervals.

Reconstructing the cited test. Fig. 4(c) of Yamagishi et al. [1] does not disclose which Bengio–Mariéthoz variant produced its matrix, so we make no cell-level attribution. With each system's non-interpolated EER threshold selected on the evaluation set, our independent-trial and shared-trial reconstructions separate the same five of six baseline pairs. The published 34-system matrix has $\binom{34}{2} = 561$ tests; the largest p among our five reconstructed results, 9.7×10^{-6}, is below even its Bonferroni threshold $.05/561 \approx 8.9 \times 10^{-5}$, while the sixth has $p = .217$. The cited derivation assumes thresholds fixed outside the test set, so these are reconstructions of its adaptation to EER rather than exact finite-sample tests.

3.4. Alternative fixed weighting rules

Within each class, every named hierarchy level receives equal conditional mass and trials divide their terminal-group mass equally. Bona-fide rules are empirical trial weights, equal mass over the three sources, and equal source→speaker mass. Spoof rules are empirical trial weights, equal mass over the five spoof strata (ASVspoof, VCC2018 HUB, VCC2018 SPO, VCC2020 Task 1, VCC2020 Task 2), equal mass over observed speaker$\times$attack cells within stratum, and equal mass successively over stratum, attack family (the attack-type field of the official 21DF trial metadata), attack and speaker. Their 3×4 crossing gives 12 positive, class-normalized rules fixed without inspecting scores. For endpoint w^1, the path from empirical weights is $w(\lambda) = (1-\lambda)w^0 + \lambda w^1$, $0 \le \lambda \le 1$. We measure its distance by maximum classwise total variation, $\max_c \frac{1}{2}\sum_i |w_{c,i} - w^0_{c,i}|$. Complete formulas, the trial-to-group maps, the simulation design and all 28 numerical bands are in the supplement (Sec. 5).

4. EXPERIMENTS

Setup. The primary set contains four author-released self-supervised systems (XLSR-Mamba [30], XLS-R+SLS [31], XLSR-Conformer [32], SSL-AASIST [33]) and four organizer-released baselines (RawNet2, LFCC-LCNN, LFCC-GMM, CQCC-GMM). We use all 533,928 evaluation-phase trials for which every score is available. Table 1 reports the conventional point results. The matched trial/PW comparison uses 5,000 replicates and seed 2026081604; seeds of the other analyses accompany their released outputs.

Primary 21DF result (Fig. 1). Both reconstructions of Fig. 4(c) separate the same five of six organizer-baseline pairs; we claim no cell-level agreement with the published matrix. The matched all-28-pair bootstrap refits the same weighted EER in every replicate. Trial resampling separates 26/28 pairs and 5/6 organizer-baseline pairs; the PW bootstrap separates 18/28 and 0/6. A constrained PW bootstrap (1,000 draws, seed 20260824) resamples speakers and attacks within each bona-fide source or spoof stratum, redraws a replicate only when a class loses all support in a stratum (7 of 1,007 attempts, all VCC2018 spoof), and then rescales each source and stratum mass to its observed value; it also separates 18/28 and 0/6. This arm was designed after the main result and is a robustness check, not prospective evidence. Thus neither threshold handling nor movement of these eight class-specific masses is necessary for the contrast. For RawNet2–CQCC-GMM, $\Delta\mathrm{EER} = -3.180$ points and the PW band is $[-9.284, 2.925]$.

Single-source SpoofCeleb check. The analysis plan, archived on 29 August 2026 while official access was pending and released with its hash in the repository of Sec. 5, fixed the four detectors, score orientation, bootstrap laws, seed and six-pair endpoint before the first successful authenticated download. On all 91,130 evaluation trials [34], trial resampling separates 6/6 pairs and the PW bootstrap separates 3/6. AASIST, SLS, SSL-AASIST and XLSR-Mamba have EERs 57.93%, 24.51%, 26.72% and 27.58%; this is an off-domain sensitivity check, not a comparison of competitive SpoofCeleb systems. Every base item has one bona-fide and nine spoof renditions under one source label, so

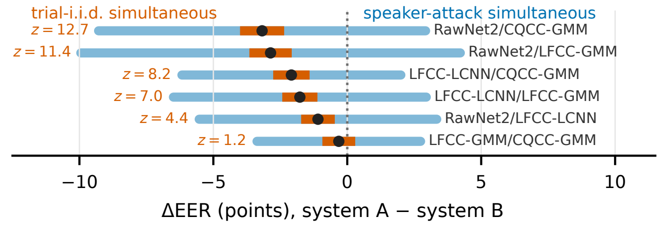

Fig. 1. The six organizer-baseline pairs, with matched all-28-pair max-t bands. Narrow: simultaneous trial-bootstrap bands; wide: simultaneous PW-bootstrap bands. The independent-trial reconstruction's z at left separates the same five pairs as the narrow bands. Dots are point estimates; no wide band excludes zero.

between-source mass cannot move; speaker and spoof-attack multiplicities still vary under PW draws. After observing this result, separate speaker-only and attack-only bootstraps separated 5/6 and 3/6; attack-only matched the PW indicator vector. The archived plan did not bind the scoring code, so a later re-scoring from freshly cloned, commit-pinned detector repositories is reproducibility evidence only: three score files were byte-identical, the fourth differed by at most 1.44×10^{-6}, and every EER and separation indicator was unchanged.

What 21DF supports. The PW bootstrap separates all 16 SSL-versus-baseline pairs, 2/6 within-SSL pairs and 0/6 within-baseline pairs. The 16 cross-cohort gaps are 19.5–23.7 points and remain separated after each leave-one-source-out refit. The two within-SSL pairs separated by the PW bootstrap (gaps 0.968 and 0.935 points) each cease to separate under one source deletion in the jackknife construction below; adjacent within-SSL gaps are only 0.032, 0.357 and 0.578 points. PW bands are 4.13–$11.36\times$ wider than matched trial bands over all pairs (median $8.14\times$).

Alternative weights. Under at least one of the 12 fixed rules, six of the 12 within-cohort pairs reverse their point ordering and none of the 16 cross-cohort pairs does; one of the 11 nonempirical linear paths reverses before maximum classwise total variation .10. Recomputing both bootstraps with each rule's weights as base weights (computed after the primary results, 5,000 draws per rule and bootstrap) separates all 16 cross-cohort pairs in all 24 rule$\times$law cells; the closest band endpoint to zero is -12.79 points. After observing these outcomes, a search over interior source and stratum masses found bona-fide masses $(.544, .244, .212)$ and spoof-stratum masses $(.133, .234, .117, .223, .294)$, in the order above and rounded, at distance .010218 from the empirical weights, that change XLSR-Mamba minus XLS-R+SLS from -0.032130 to $+0.000699$ EER points. This is a found upper bound, not a minimum, and is too small to imply practical superiority. The search found no cross-cohort reversal, which is not a proof that none exists.

Table 1. Point EER (%) on the 21DF evaluation set. "Release" identifies the score-file source. Rank is point order only, not an uncertainty claim.

System	Release	EER (%) ↓	Rank
XLSR-Mamba	author	1.884	1
XLS-R+SLS	author	1.916	2
XLSR-Conformer	author	2.273	3
SSL-AASIST	author	2.852	4
RawNet2	organizer	22.383	5
LFCC-LCNN	organizer	23.477	6
LFCC-GMM	organizer	25.247	7
CQCC-GMM	organizer	25.563	8

Alternative released scores on 21DF. On the eleven complete Speech DF Arena score files for the same 533,928 trials, trial and PW bootstraps separate 52/55 and 38/55 pairs; the median ratio of PW to trial bootstrap standard deviations over the 55 pairs is 8.45. These are descriptive results for that separate score collection, not population confidence intervals.

Source deletions. The bona-fide sources ASVspoof, VCC2018 and VCC2020 contain 67, 12 and 14 speakers. Source deletion uses a third construction: with delete-one speaker, attack and observed-cell jackknife variances V_s, V_a and V_{sa} of each contrast, the interval is $\hat{\Delta} \pm q\sqrt{\max(V_s + V_a - V_{sa}, V_s, V_a, 0)}$, where q is a Gaussian max-t critical value recomputed over all 28 contrasts for each remaining subset (200,000 draws). On the full data this construction separates 18/28 pairs (16 cross-cohort, 2 within-SSL, 0 within-baseline), the same verdicts as the PW bootstrap. Leave-one-source-out refits preserve separation for all 16 cross-cohort pairs, remove both separated within-SSL pairs and separate three within-baseline pairs in at least one deletion. Thus cross-cohort separation survives these source changes, while some within-cohort separation decisions change with them.

ASVspoof 5 check. For SSL-AASIST, AASIST, XLS-R+SLS and XLSR-Mamba on all 680,774 Track 1 trials [3], resampling the 367 target speakers, the 370 bona-fide-only speakers and the 16 attacks as three independent multinomial draws changes the separated count from 6/6 under trial resampling to 5/6 and gives a median simultaneous-band width ratio of $27.27\times$; both criteria (at least one lost separation, median ratio at least 2) were fixed in the released plan before scoring completed. This check is intentionally EER-only because it transfers the paper's ΔEER comparison; ASVspoof 5's primary Track 1 metric is minDCF. It is not a system-performance replication or population claim.

Coverage boundary. The candidates are the raw exact-cell inclusion–exclusion interval $\hat{\Delta} \pm 1.96\sqrt{V_s + V_a - V_{sa}}$ with the jackknife variances defined above, its variance-floor version using $\max(V_s + V_a - V_{sa}, V_s, V_a)$, and the 2.5–97.5 percentile PW band. We simulated 48 bivariate random-effects models on the 21DF trial-to-group structure: two regimes fitted to the RawNet2/LFCC-LCNN and XLSR-Mamba/XLS-R+SLS pairs, four interaction shares (0, .25, .50, .75: the fraction of the fitted spoof speaker-plus-attack variance moved into a speaker×attack interaction, total held fixed), Gaussian or t_5 effects, and target Δ of 0, 0.5 and 2 points, 1,000 replicates per cell (parameters in the supplement). Over the 24 low-EER cells the minimum nominal-95% coverages are .845, .845 and .855; all three are below .90 in the same 11/24 cells. This simulation evaluates procedures under those imposed models; it does not establish that either model describes 21DF.

5. DISCUSSION

Reporting and limits. The complete 28-row band table, trial-to-source/stratum map, weighting formulas, 48-cell simulation parameters, the SpoofCeleb and ASVspoof 5 analysis plans with their hashes, and all artifact identifiers are in `paper/SUPPLEMENT.md` at `github.com/rvirgilli/asvspoof2021-df-clustered-uncertainty`, tag `icassp2027-submission`. The legacy SSL-AASIST/AASIST files lack their originating run logs. Benchmark reports should publish trial-to-speaker/attack membership and source weights, not only trial counts. A band containing zero means the procedure does not separate that pair; it does not establish equality. We make no population, rank-confidence or formal Arena-inference claim.

6. CONCLUSION

On fixed released scores, matched 21DF bootstraps change the separated organizer-baseline count from 5/6 to 0/6, and the single-source SpoofCeleb check, on a different four-detector roster, changes it from 6/6 to 3/6. Because SpoofCeleb has one source, between-source movement cannot explain the latter contrast; an attack-only bootstrap computed after the result matches its 3/6 indicator vector. All 16 pairs between the four SSL detectors and the four organizer baselines remain separated under both bootstraps, each leave-one-source-out refit of the declared jackknife construction and all 12 fixed weighting rules, and none reverses its point ordering in the searched directions; within-cohort separation changes with the resampling procedure and with source deletion, and within-cohort point ordering reverses under some weighting rules. Population confidence requires a defensible acquisition model and an inferential procedure valid under its dependence structure.

7. REFERENCES

[1] Junichi Yamagishi, Xin Wang, Massimiliano Todisco, et al., "ASVspoof 2021: accelerating progress in spoofed and deep-

fake speech detection," in *Proc. ASVspoof Challenge Workshop*, 2021, pp. 47–54.

[2] Xuechen Liu, Xin Wang, Md Sahidullah, et al., "ASVspoof 2021: towards spoofed and deepfake speech detection in the wild," *IEEE/ACM Trans. Audio, Speech, Language Process.*, vol. 31, 2023.

[3] Xin Wang, Héctor Delgado, Nicholas Evans, et al., "ASVspoof 5: evaluation of spoofing, deepfake, and adversarial attack detection using crowdsourced speech," *IEEE Trans. Audio, Speech, Language Process.*, 2026, arXiv:2601.03944.

[4] Sandipana Dowerah, Atharva Kulkarni, Ajinkya Kulkarni, et al., "Speech DF Arena: a leaderboard for speech deepfake detection models," *arXiv:2509.02859*, 2025.

[5] Hashim Ali, Nithin Sai Adupa, et al., "A SUPERB-style benchmark of self-supervised speech models for audio deepfake detection," in *Proc. ICASSP*, 2026.

[6] Héctor Delgado, Massimiliano Todisco, Md Sahidullah, et al., "ASVspoof 2017 Version 2.0: meta-data analysis and baseline enhancements," in *Proc. Odyssey*, 2018, pp. 296–303.

[7] Samy Bengio and Johnny Mariéthoz, "A statistical significance test for person authentication," in *Proc. Odyssey*, 2004.

[8] Octavian Pascu, Adriana Stan, Dan Oneata, Elisabeta Oneata, and Horia Cucu, "Towards generalisable and calibrated audio deepfake detection with self-supervised representations," in *Proc. Interspeech*, 2024, pp. 4828–4832.

[9] Yuankun Xie, Haonan Cheng, Jiayi Zhou, Xiaoxuan Guo, Tao Wang, Changhao Zhang, Jian Liu, Weiqiang Wang, Ruibo Fu, Xiaopeng Wang, Hengyan Huang, Xiaoying Huang, Long Ye, and Guangtao Zhai, "AT-ADD: a benchmark and challenge for robust and all-type audio deepfake detection," *arXiv:2608.23437*, 2026.

[10] Xin Wang and Junichi Yamagishi, "A comparative study on recent neural spoofing countermeasures for synthetic speech detection," in *Proc. Interspeech*, 2021.

[11] Norman Poh, Alvin Martin, and Samy Bengio, "Performance generalization in biometric authentication using joint user-specific and sample bootstraps," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 29, 2007.

[12] Jin Chu Wu, Alvin F. Martin, et al., "The impact of data dependence on speaker recognition evaluation," *IEEE/ACM Trans. Audio, Speech, Language Process.*, vol. 25, 2017.

[13] Jin Chu Wu, Alvin F. Martin, Craig S. Greenberg, et al., "Significance test in speaker recognition data analysis with data dependency," Tech. Rep. NISTIR 7884, National Institute of Standards and Technology, 2012.

[14] Ruud M. Bolle, Nalini K. Ratha, et al., "Error analysis of pattern recognition systems—the subsets bootstrap," *Comput. Vis. Image Underst.*, vol. 93, pp. 1–33, 2004.

[15] Riccardo Fogliato, Pratik Patil, et al., "Confidence intervals for error rates in 1:1 matching tasks," *Int. J. Comput. Vis.*, 2024.

[16] A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller, "Robust inference with multiway clustering," *J. Bus. Econ. Stat.*, vol. 29, no. 2, pp. 238–249, 2011.

[17] Peter McCullagh, "Resampling and exchangeable arrays," *Bernoulli*, vol. 6, 2000.

[18] Art B. Owen, "The pigeonhole bootstrap," *Ann. Appl. Stat.*, vol. 1, 2007.

[19] Kylie Anglin, "Estimating uncertainty in classifier performance with applications to large language models and nested data," *arXiv:2606.26422*, 2026.

[20] Maximilian Bisani and Hermann Ney, "Bootstrap estimates for confidence intervals in ASR performance evaluation," in *Proc. ICASSP*, 2004.

[21] Zhe Liu and Fuchun Peng, "Statistical testing on ASR performance via blockwise bootstrap," *arXiv:1912.09508*, 2019.

[22] Zexin Zhuang, Yanhang Li, and Zhichao Fan, "Pre-registering the detectable effect: a paired-MDE budget for 4-bit quantization benchmarks, with a pilot audit," *arXiv:2605.28873*, 2026.

[23] Chin Yuen Kwok et al., "Bona fide cross testing reveals weak spot in audio deepfake detection systems," in *Proc. Interspeech*, 2025, arXiv:2509.09204.

[24] Héctor Delgado, Giorgio Ramondetti, Emanuele Dalmasso, Gennady Karvitsky, Daniele Colibro, and Haydar Talib, "On deepfake voice detection—it's all in the presentation," in *Proc. ICASSP*, 2026.

[25] Samuel Pagon, Yixuan Shen, et al., "What do deepfake benchmarks measure? An audit using frozen self-supervised representations," *arXiv:2606.26384*, 2026.

[26] Vojtěch Staněk, Eva Trnovská, Kamil Malinka, Anton Firc, et al., "Ethical and technical limits of deepfake speech datasets," in *Proc. Interspeech*, 2026.

[27] Daniyal Kabir Dar and Arun Ross, "Probing speaker identity sensitivity in audio deepfake detectors," *arXiv:2607.21820*, 2026.

[28] Diaa Al Mohamad, Jelle J. Goeman, et al., "Simultaneous confidence intervals for ranks," *arXiv:1812.05507*, 2019.

[29] José Luis Montiel Olea and Mikkel Plagborg-Møller, "Simultaneous confidence bands: theory, implementation, and an application to SVARs," *J. Appl. Econom.*, vol. 34, 2019.

[30] Yang Xiao and Rohan Kumar Das, "XLSR-Mamba: a dual-column bidirectional state space model for spoofing attack detection," *IEEE Signal Process. Lett.*, 2025.

[31] Qishan Zhang, Shuangbing Wen, and Tao Hu, "Audio deepfake detection with self-supervised XLS-R and SLS classifier," in *Proc. ACM Multimedia*, 2024, pp. 6765–6773.

[32] Eros Rosello et al., "A conformer-based classifier for variable-length utterance processing in anti-spoofing," in *Proc. Interspeech*, 2023.

[33] Hemlata Tak et al., "Automatic speaker verification spoofing and deepfake detection using wav2vec 2.0 and data augmentation," in *Proc. Odyssey*, 2022.

[34] Jee-Weon Jung, Yihan Wu, Xin Wang, Ji-Hoon Kim, Soumi Maiti, Yuta Matsunaga, Hye-Jin Shim, Jinchuan Tian, Nicholas Evans, Joon Son Chung, Wangyou Zhang, Seyun Um, Shinnosuke Takamichi, and Shinji Watanabe, "SpoofCeleb: Speech deepfake detection and SASV in the wild," *IEEE Open Journal of Signal Processing*, vol. 6, pp. 68–77, 2025.